METRIC

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SUPERSEDING
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FEDERAL SPECIFICATION

STANDARD TEST FLUIDS, HYDROCARBON

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers standard test fluids suitable for contact testing of various materials to determine their resistance to hydrocarbon fluids (see 6.1).

1.2 Classification. Hydrocarbon test fluids shall be of the following types, as specified (see 6.2).

Type I	- 2,2,4-Trimethylpentane.
Type II	- 2,2,4-Trimethylpentane and toluene mixture (60:40).
Type III	- 2,2,4-Trimethylpentane and toluene mixture (70:30).
Type IV	- Petroleum base oil, low swell characteristic.
Type V	- Petroleum base oil, medium swell characteristic.
Type VI	- Petroleum base oil, high swell characteristic.
Type VII	- Hydrocarbon mixture with added sulfur compounds.
Type VIII	- JP-4 simulant.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

TT-T-548	- Toluene, Technical.

Federal Standards:

FED-STD-102	- Preservation, Packaging, and Packing Levels.
FED-STD-313	- Material Safety Data Sheets, Preparation and the Submission of.
FED-STD-601	- Rubber: Sampling and Testing.
FED-STD-791	- Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal specification, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Auburn, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-I-27686	- Inhibitor, Icing, Fuel Systems.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-290	- Packaging, Packing, and Marking of Petroleum and Related Products.

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Federal Regulations:

29 CFR Part 1910.1200, Hazard Communication.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the superintendent of Documents, U.S. Government Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Chemical Society (ACS):

Reagent Chemicals, American Chemical Society Specifications

(Application for copies should be addressed to the American Chemical Society, 1155 16th Street, N.W., DC 20036.)

American Society for Testing and Materials (ASTM):

A 2	- Reference Materials and Blending Accessories, Annex 2 to Motor Fuels,Section I, Volume 05.04.
D 92	- Flash and Fire Points by Cleveland Open Cup.
D 362	- Industrial Grade Toluene.
D 445	- Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity).
D 471	- Rubber Property — Effect of Liquids.
D 611	- Aniline Point and Mixed Aniline Point of Petroleum Products and Hydrocarbon Solvents.
D 4176	- Free Water and Particulate Contamination in Distillate Fuels (Clear and Bright Pass/Fail Procedure).
E 260	- General Gas Chromatography Procedures.
E 355	- Recommended Practice for Gas Chromatography Terms and Relationships.

(The standards listed above are included in Volumes 05.01, 05.03, 05.04, 09.01, 14.01, and 15.05 of the Annual Book of ASTM Standards. The methods can also be purchased separately. Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. With one exception, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. The exception is that 29 CFR Part 1910.1200 shall take precedence over the text of this specification.

3. REQUIREMENTS

3.1 Materials. Standard test fluids shall consist of hydrocarbons and other compounds as specified herein, and as necessary to produce fluids meeting the requirements of this specification. The test fluids shall be furnished in unit containers of the sizes and types specified (see 6.2).

3.1.1 tert-Butyl disulfide. tert-Butyl disulfide shall be of 98 percent minimum purity. (Warning! Stench. Causes shin and eye irritation.)

3.1.2 Cyclohexane. Cyclohexane shall conform to Reagent ACS Grade. Other grades of cyclohexane are also acceptable, provided they are of 99 percent minimum purity. (DANGER! Extremely flammable.)

3.1.3 Decahydronaphthalene. Decahydronaphthalene (also commonly referred to as decalin) shall be of 97 percent minimum purity, and may contain mixed cis and trans isomers in any proportion. (CAUTION! Combustible. WARNING! Harmful if inhaled. Harmful if absorbed through skin. Causes skin and eye irritation.)

3.1.4 2-Methoxyethanol. 2-Methoxyethanol (also commonly referred to as ethylene glycol monomethyl ether) shall conform to MIL-I-27686. (CAUTION! Combustible. WARNING! Harmful if inhaled. Harmful if absorbed through skin. Can cause blood disorders. Can cause nervous system injury. Can cause adverse reproductive effects.)

3.1.5 1-Methylnaphthalene. 1-Methylnaphthalene shall be of 95 percent minimum purity. (WARNING! Harmful if inhaled. Causes skin and eye irritation.)

3.1.6 2-Methyl-2-propanethiol. 2-Methyl-2-propanethiol (also commonly referred to as tert-butyl mercaptan) shall be of 95 percent minimum purity. (DANGER! Extremely flammable. WARNING! Causes skin and eye irritation.)

3.1.7 Tetrahydronaphthalene. Tetrahydronaphthalene (also commonly referred to as tetralin) shall be of 97 percent minimum purity. (WARNING! Causes skin and eye irritation.)

3.18 Toluene. Toluene shall conform to TT-T-548, Reagent ACS Grade, ASTM D 362, or to ASTM Reference Fuel Grade Toluene as specified in Annex A2, Motor Fuels, Section I, Volume 05.04 of the Annual Book of ASTM Standards. (WARNING! Flammable. Harmful if inhaled. Harmful if absorbed through skin. Causes skin and eye irritation.)

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3.1.9 2,2,4-Trimethylpentane. 2,2,4-Trimethylpentane (also commonly referred to as isooctane) shall conform to ASTM Knock Test Reference Fuel Isooctane, Annex A2, Motor Fuels, Section I, Volume 05.04 of the Annual Book of ASTM Standards; or to Reagent ACS Grade. (WARNING! Flammable.)

3.2 Composition of types I, II, and III fluids. The composition of types I, II, and III fluids shall be as specified in table I.

TABLE I. Composition of types I, II, and III fluids.

Ingredient material	Volume percent		
	Type I	Type II	Type III
Toluene	—	40	30
2,2,4-Trimethylpentane	100	60	70

3.3 Properties of types IV, V, and VI fluids. Types IV, V, and VI fluids shall be petroleum base oils having the properties specified in table II.

TABLE II. Properties of types IV, V, and VI fluids.

Property	Type IV	Type V	Type VI
Aniline Point, °C	124.0 ±1.0	93.0 ±3.0	70.0 ±1.0
Kinematic viscosity, mm^2/s			
min.	18.7 1/	19.2 1/	31.9 2/
max.	21.0 1/	21.5 1/	34.1 2/
Flash point, min, °C	243	240	163

1/ Measured at 99 °C.
2/ Measured at 37.8 °C.

3.4 Composition of types VII and VIII fluids. The composition of types VII and VIII fluids shall be as specified in table III.

TABLE III. Composition of types VII and VIII fluids.

Ingredient material	Volume percent	
	Type VII	Type VIII
tert-Butyl disulfide	0.985	—
Cyclohexane	59.0	25.0
Decahydronaphthalene	—	5.0
2-Methoxyethanol	—	0.3 1/
1-Methylnaphthalene	—	1.7

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TABLE III. Composition of types VII and VIII fluids. (Cont'd)

Ingredient material	Volume percent	
	Type VII	Type VIII
2-Methyl-2-propanethiol	0.015	—
Tetrahydronaphthalene	—	3.0
Toluene	30.0	15.0
2,2,4-Trimethylpentane	10.0	50.0

1/ 2-Methoxythanol is specified at twice the concentration normally found in JP-4 to enhance the efforts of this chemical on materials under test. Normal concentration in JP-4 is 0.10 to 0.15 volume percent.

3.5 Material Safety Data Sheets. Material Safety Data Sheets shall be prepared for the test fluids in accordance with FED-STD-313 (see 6.5).

3.6 Workmanship. The finished standard test fluids shall be visually free from undissolved water, sediment, and suspended matter; and shall be clear and bright.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classed as follows:

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a. Quality conformance inspection (see 4.4).
b. Inspection of preparation for delivery (see 4.6).

4.3 Inspection lot.

4.3.1 Bulk lot. A bulk lot is defined as an indefinite quantity of a single homogeneous test fluid offered for acceptance in a single isolated container, manufactured as a single isolated batch, or manufactured as a single plant run (not exceeding 5 days) through the same processing equipment with no change in the ingredient materials.

4.3.2 Packaged lot. A packaged lot is defined as an indefinite number of drums or smaller unit packages of identical size and type of fluid, offered for acceptance, and filled with a homogeneous mixture of material from a bulk lot.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Sampling for the inspection of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105, at inspection level II, with an AQL of 1.0 percent defective.

4.4.1.2 Sampling for tests. Each packaged lot shall be sampled for testing in accordance with MIL-STD-105.

4.4.2 Inspection. Inspection procedures shall be in accordance with FED-STD-791, method 9601.

4.4.2.1 Examination of filled containers. Samples selected in accordance with 4.4.1.1 shall be examined for compliance with MIL-STD-290 with regard to closure, sealing, leakage, and marking.

4.4.2.2 Tests. Samples selected in accordance with 4.4.1.1 shall be tested as specified in 4.5.

4.5 Tests.

4.5.1 Purity of ingredient materials. Materials listed in 3.1.1 through 3.1.9 which conform to cited specifications (cyclohexane, 2-methoxyethanol, toluene, and 2,2,4-trimethylpentane) shall be tested in accordance with the applicable specification. Materials listed in 3.1.1 through 3.1.9 which specify only a minimum purity (tert-butyl disulfide, decahydronaphthalene, 1-methylnaphthalene, 2-methyl-2-propanethiol, and tatrahydronaphthalene) shall be tested for purity by a gas chromatographic assay conforming to ASME E 260 and E 355. The actual test procedures used by the contractor or the contractor's chemical supplier shall be available to the Government upon request.

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4.5.2 Composition of types I, II, III, VII, and VIII test fluids. The composition of type I, II, III, VII, and VIII fluids is governed primarily by the accuracy in measuring the volumes of the ingredient materials composing the fluid Target volumes for each ingredient material shall be calculated by multiplying the volume percentage from table I or table III times the volume of test fluid to be prepared. The actual volume of each ingredient material composing the fluid shall be within ±1.0 percent of the calculated target volume.

4.5.3 Properties of types IV, V, and VI test fluids. The properties of types IV, V, and VI test fluids shall be tested by the methods specified in table IV.

TABLE IV. Test methods.

Test	ASTM Method
Aniline Point	D 611
Kinematic viscosity	D 445
Flash Point	D 92

4.5.4 Workmanship. The finished standard test fluids shall be tested in accordance with method A or B of ASTM D 4176 at the ambient temperature or 25 °C.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of preparation for delivery.

4.6.1.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Inspection lot. The inspection shall be as defined in 4.3, packed for shipment.

4.6.1.3 Sampling. Sampling for examination of preparation for delivery shall be selected at random from each inspection lot in accordance with MIL-STD-105.

4.6.1.4 Examination. Samples selected in accordance with 4.6.1.3 shall be examined for the defects listed below. The AQL shall be 1.0 percent defective.

Number	Defect	Paragraph
101.	Unit containers not of size or style specified.	3.1
102.	Material Safety Data Sheet not prepared in accordance with FED-STD-313.	3.5
103.	Unit containers not in accordance with the requirements of MIL-STD-290.	5.1
104.	Exterior containers, when required, not in accordance with the requirements of MIL-STD-290.	5.1
105.	Marking not in accordance with MIL-STD-290.	5.1
106.	Special marking for types I, II, III, VII, and VIII not as specified.	5.2.1

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107.	Special marking for types IV, V, and VI not as specified.	5.2.2
108.	Palletization, when specified (see 6.2), not in accordance with MIL-STD-290.	5.3

5. PREPARATION FOR DELIVERY

5.1 Preservation/packaging, packing and marking. The unit containers of the size and styles specified in 3.1 (see 6.2), the exterior containers required for the unit containers, and the marking of such unit and exterior containers shall be in accordance with the applicable requirements of MIL-STD-290. The level of protection shall be level B, level C or commercial as specified (see 6.2) and shall comply with the requirements of MIL-STD-290. Civil agencies should refer to FED-STD-102 for definitions and applications of the various levels of protection for supplies and equipment.

5.2 Special marking. The following information shall be legibly marked on each unit container and, as applicable, each exterior container.

5.2.1 Types I, II, III, VII, and VIII fluids. The special markings for types I, II, III, VII, and VIII shall be as follows:

!DANGER!
EXTREMELY FLAMMABLE.
VAPORS HARMFUL IF INHALED
VAPORS MAY CAUSE FLASH FIRE

Keep away from heat, sparks, and open flame.
Vapors are heavier than air and may gather in low places, resulting in explosion hazard.
Keep container closed.
Use adequate ventilation.
Avoid build-up of vapors.
Avoid prolonged breathing of vapor ar spray mist.
Avoid prolonged or repeated skin contact.

5.2.2 Types IV, V, and VI fluids. The special marking for types IV, V, and VI shall be as follows:

Avoid prolonged or repeated skin contact.

5.3 Palletization. When specified (see 6.2), the packed fluid shall be palletized in accordance with MIL-STD-290.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The standard test fluids in this specification are intended for use in testing paint, elastomers, rubber, and various other materials for resistance to hydrocarbon fluids. Type VIII test fluid is intended to simulate a typical JP-4 aviation turbine fuel conforming to MIL-T-5624.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of test fluid required (see 1.2).
c. Quantity of test fluid required. The unit of purchase is one U.S. gallon (3.785 liters) at 60 °F (15.6 °C).
d. Type and size of containers required (see 5.1).
e. Level of protection required (see 5.1).
f. When palletization is required (see 5.3).

6.3 Cross-reference of test fluids. Types I through VI test fluids in this specification are equivalent to various fluids specified in ASTM D 471 and in FED-STD-601, method 6001. Types VII and VIII, however, are unique to this specification. The cross-reference relationships are shown in table V.

TABLE V. Test fluid cross-reference.

TT-S 735	ASTM D 471	FED-STD-601, method 6001
Type I	Reference Fuel A	Medium No. 4
Type II	Reference Fuel D	—
Type III	Reference Fuel B	Medium No. 6
Type IV	Oil No. 1	Medium No. 1
Type V	Oil No. 2	Medium No. 2
Type VI	Oil No. 3	Medium No. 3
Type VII	—	—
Type VIII	—	—

6.4 Hazardous materials. This specification may involve the use of hazardous materials, and does not purport to address all of the safety problems associated with their use. It is the responsibility of whoever uses this specification to consult and establish appropriate safety and health practices, and to determine the applicability of regulatory limitations prior to use.

6.5 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in Appendix B of FED-STD-313. One copy should also be provided to the procuring activity.

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6.6 Subject term (key word) listing.

Hydrocarbons
Test fluids

MILITARY INTERESTS:
ACTIVITIES

Custodians:
Army - ME
Navy - SH
Air Force - 20

Review Activities:
Army - EA, AT, AV
Navy - AS
Air Force - 68
DLA - GS

User Activity:
Army - AR
Navy - OS

CIVIL AGENCY COORDINATING

VA - OSS
NASA - MSF
GSA -FSS

PREPARING ACTIVITY:
Army - ME

Project 9150-0708

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
TT-S-735A	Standard Test Fluids, Hydrocarbon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREA

a. Paragraph Number and Wording:

a. Recommended Wording:

b. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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